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Practical Computational Methods for Social Scientists

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Fundamentals

Outline



1. Fundamental concepts of text analysis:
 1. Text as data
 2. Text as social science data
 3. Demystifying concepts I
 4. Demystifying concepts II
2. Exercise

Text as data



Text is the new frontier of:

- Data
- Methods
- Social Science (Spirling, 2022)

In terms of data, we inhabit an era of voluminous, readily-accessible text.

In terms of methods, we can harness a range of mature (e.g., sentiment analysis, topic modelling) and novel (e.g., word embeddings, LLMs) methods that can make sense of voluminous, unstructured text.

In terms of social science, we can answer existing questions at scale / granularity or new questions for the first time.

Text as data



Types of methods:

- **Descriptive inference:** how to characterise text; *vector space model*, *bag of words*, *(dis)similarity measures*, diversity, complexity, style, bursts.
- **Supervised techniques:** dictionaries, *sentiment analysis*, categorising.
- **Unsupervised techniques:** cluster analysis, PCA, *topic modelling*, embeddings. (Spirling, 2022)

Text as social science data



Text analysis in social science research (Grimmer et al., 2022) :

1. Representation = from high-dimensional to low-dimensional
2. Discovery = useful ways of conceptualising and organising text
3. Measurement = describing text in an accurate and insightful manner
4. Inference = making predictions and causal claims

Grimmer et al. (2022) propose a common research trajectory for text analysis in the social sciences. It is sequential but iterative (e.g., where measures are refined or jettisoned depending on the results of the discovery phase).

All four phases may not be part of every project but 1 and 2 are foundational.

Text as social science data



Representation is all about how we reduce text from a *high-dimensional* to a *low-dimensional* state.

High dimensional = lots of complexity or aspects to the text

Low dimensional = little complexity or few aspects to the text

Text as social science data



<i>This morning on the harbour</i>	5
<i>When I said goodbye to you</i>	6
<i>I remember how I swore</i>	5
<i>That I'd come back to you one day</i>	8
<i>And as the sunset came to meet the evening on the hill</i>	12
<i>I told you I'd always love you</i>	7
<i>I always did and I always will</i>	7

The Body of an American (2001) by The Pogues.

Text as social science data



Discovery is all about how we conceptualise and identify the aspects of the text that are relevant to our research question.

Can be a data-driven process (e.g., clusters) but we can also theorise based on prior work or intuition / substantive expertise.

In essence, what are the patterns or structures in the data we want to reveal?

Returning to our lyrical example, is there a pattern or structure to The Pogues songs? Do they cover similar or diverse themes? Are they structured in a similar way (e.g., number of verses and choruses)?

Text as social science data



Measurement is all about how we describe the prevalence of our concepts / aspects of the text that are relevant to our research question.

We can adopt standard measures and metrics e.g., cosine similarity and apply these across our texts.

Measurement requires validation!

Text as social science data



The places mentioned in "The Body of an American" by The Pogues are:

- New York City
- Boston
- PA (Pennsylvania)
- Pittsburgh
- Amerikay (a traditional Irish term for America)
- Spain (implied by "Spanish wine from far away")
- USA

For example, if we look at the full lyrics for the song The Body of an American, there are seven places mentioned.

Text as social science data



Inference is all about how we take our measures and make predictions or causal claims about social phenomena.

Did a change in X (e.g., tone of policy announcement) produce a change in Y (e.g., public support for policy)?

Demystifying concepts I



Document = a single unit of text that is being analysed e.g., a paragraph, article, report, book, speech, law etc.

Corpus = a collection of documents used for text analysis e.g., a dataset containing activity descriptions of overseas charities.

Corpora = a collection of collections e.g., datasets of activity descriptions from multiple charity jurisdictions.

A document is often the unit of analysis in text analysis research.

Demystifying concepts II



Type = a unique unit of text in a document or corpus. Often a word but can also be other meaningful sequences of characters e.g., numbers. The set of types is called a *vocabulary*.

Term = similar to a type but also including units of text that do not appear in the corpus but are generated or inferred e.g., stems and lemmas.

Token = a particular instance of a term in a corpus.

Demystifying concepts II



This morning on the harbour

When I said goodbye to you

I remember how I swore

That I'd come back to you one day

And as the sunset came to meet the evening on the hill

I told you I'd always love you

I always did and I always will

**How many types
are in the
document?**

**How many tokens
are in the
document?**

32 types

50 tokens

However “I” and “I’d” are counted separately. But “And” and “and” are counted together.

Exercise

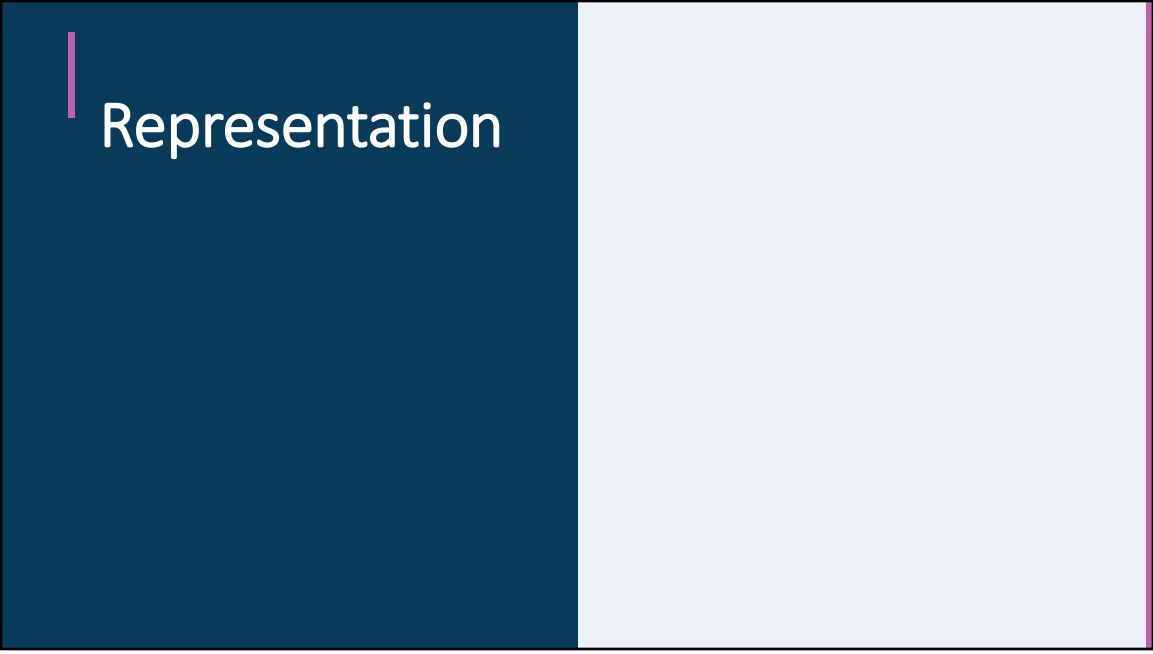


Overseas Charities

Using the file “*acnc-overseas-activities-2022.csv*”, read the activity statements of 15 charities: 5 small organisations; 5 medium organisations; 5 large organisations.

Analyse the text as follows:

1. Is there a structure or pattern in these statements?
2. How many unique words are there in each text / in total?
3. How many terms and tokens are there?



Representation

Outline

1. Converting text to numeric:
 1. Workflow
 2. Pre-processing
2. Simple representations of text:
 1. Bag of words
 2. DFM / DTM
3. Other considerations:
 1. Weighting
 2. N-grams

Converting text to numeric



Decide what constitutes a document (needs to be machine readable).

Remove superfluous material (e.g., capitalisation, punctuation, non-alphanumeric characters).

Separate document into useful elementary pieces i.e., tokens.

Adding descriptive annotations that preserve context e.g., tagging.

Map tokens to a common form i.e., stemming/ lemmatisation.

Perform analysis (Spirling, 2022).

Not all of these steps are necessary for every use of text analysis, and what you do within each step may vary also (e.g., stemming vs lemmatisation). In addition, the order of these steps is your decision!

Converting text to numeric



1. Choose unit of analysis.
2. Tokenise.
3. Reduce complexity:
 1. Convert to lowercase
 2. Remove punctuation
 3. Remove stop words
 4. Create equivalence classes (lemmatisation / stemming)
 5. Filter by frequency
4. Construct Document-Feature Matrix (Grimmer et al., 2022).

This workflow or recipe is a more technical implementation of the general workflow proposed by Spirling (2022). The order also matters, as we will see in the practical. We will look at each of these steps in turn during the practical – for now let's focus on what this recipe produces.

Converting text to numeric



The recipe outlined in the previous slide produces a “bag of words” representation: a count of how many times each term appears in a document.

What do we gain and lose from this representation?

Even if the bag of words is not the objective of the analysis, it is a necessary part of a larger analysis e.g., topic modelling (Spirling, 2022).

Using the bag of words recipe means we are not interested in (or able to analyse) word order.



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Representing text as numeric



A single document represented as a row of counts of term frequencies = **vector space model**.

victims	viet	vietnam	village	virginia	vision	volunteer	war	work	works	zoom
1	2	2	1	1	1	1	1	1	3	1

If we stack rows we get a **Document Term Matrix (DTM)**.

A vector space model represents documents as a sequence of term counts. This representation is useful as we can not only compare the raw counts but also the sequence.

Word clouds and other visualisations are nice but the actual underpinning data structure / format is what's called a Document Term Matrix / Document Feature Matrix. The features of the document are the terms found in it and the cells capture counts of those terms in the document.

Other considerations: Weighting



Term Frequency – Inverse Document Frequency (TF-IDF)

The core idea is to prioritise highly frequent words in the document but are rare in the corpus overall.

$$W_{ij} = W_{ij} * \log \frac{N}{n_j}$$

Rare words are given larger weights, common words are given smaller / zero weights).

You may wonder why we adjust the raw counts of terms in the corpus? The idea is to aid our analysis by a) ensuring documents that use the same terms but in very different frequencies are still considered similar and b) improve our ability to discriminate between documents i.e., easier to identify documents that use particular words that are rare overall – good for information retrieval tasks.

Other considerations: N-grams



Usually we tokenise text by dividing it into subunits based on single terms.

“Text analysis is transforming social science.”

['analysis', 'science', 'social', 'text', 'transforming']

['analysis transforming', 'social science', 'text analysis', 'transforming social']

['analysis transforming social', 'text analysis transforming', 'transforming social science']

By tokenising text we create unigrams: each term is treated separately.

Bigrams and trigrams can often be often important depending on your analysis e.g., “social” and “science” will not be linked through a stem or lemma but they are substantively linked. The problem with bigrams and trigrams (and higher) is they can be computationally intensive to process.

When might you use bigrams and trigrams in your analysis? If you want to preserve word order or multiword terms.

Conclusion:

<https://github.com/SGSSOnline/text-analysis/tree/main>

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